



Prostate cancer theragnostics biomarkers: An update

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Biomarkers are molecules such as proteins, genes, or other substances that may be tested to determine the stage of the tumor in a patient. The role of prostate cancer biomarkers is pivotal and the combination of prostate cancer immunotherapy with efficient biomarkers has emerged as a beneficial treatment strategy and its use has increased rapidly. The two primary objectives of this current prostate cancer early detection programs were recognizing non-symptomatic individuals with prostate cancer requiring prostatic core biopsy and identifying men with prostate cancer who might benefit from definitive medical treatment. The progress that has been made so far in the identification of the biomarkers that can be used for the classification, prediction and prognostication of prostate cancer, and as major targets for its clinical intervention has been well summarized in this review.

Keywords: Exosomes; Immunotherapy; Liquid biopsies; Metastasis; Prostatectomy

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INCIDENCE

Prostate cancer is a disease with several clinical and molecular features [1]. It is the most prevalent type of cancer in men in over half of the world's countries, and the leading cause of cancer mortality in 46 of those countries, with the Caribbean holding the highest rates [2]. Prostate cancer is the second prevalent cancer and is the fifth-leading cause of death worldwide, accounting 1.3 million new cases and 359,000 deaths in 2018 [3]. It is also one of the most frequent cancers among men in US (about 174,650 new cases in 2019), almost 20% of all newly diagnosed cancers [4]. Prostate cancer affects one in every six American males at some point in their lives. Of them, only one may die from the disease, and

majority of them may not show any clinical symptoms [5]. In comparison, White men are 55% more likely to receive a prostate cancer diagnosis than Asian-American men; yet, research have shown that Asian-American men are more likely to develop high-grade prostate cancer [6]. It is anticipated that 20% of individuals with prostate cancer have a high-risk malignancy that might advance to a potentially fatal condition, despite the fact that the majority of patients have been diagnosed with inactive or extremely slow fostering disease [7]. Advanced prostate cancer is still completely incurable, even though the majority of cases are found in their early, treatable stages. Patients with prostate cancer who had metastatic progression experienced a five-year survival rate of 30.6% between 2011 and 2017 [8]. Ageing, ances-

Received: 9 July, 2024 • **Revised:** 2 September, 2024 • **Accepted:** 10 October, 2024 • **Published online:** 29 October, 2024

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tral history, and genetic predisposition plays a major role in the development of prostate cancer [9]. Globally, prostate cancer ranks second overall in terms of incidence and is one of the top five causes of cancer-related mortality in men; however, in Western nations, it is the second most common cause of tumor-related fatality in men [10]. One of the most prevalent forms of cancer among men in developed nations is localized prostate cancer, and its prevalence is rising [11]. Approximately twenty million individuals received cancer diagnosis in 2020, and 10 million cancer-related deaths were predicted globally. Men's lifespan is predicted to extend with improvements made in prostate cancer treatment and diagnosis in patients [12]. By 2040, there would be 2.9 million new cases of prostate cancer yearly, up from 1.4 million in 2020, according our statistical predictions. Large rises in the disease are expected to be driven by increased life expectancy and changing age patterns [13]. Familial history, age as well as ethnic background are recognized as a risk factor for prostate cancer. For individuals with a family history of the illness in first-degree relatives, their relative risk of prostate cancer rises about two to three times [14]. Families with two first-degree family members diagnosed with prostate cancer at any point in their life or one first-degree relative and two second-degree family members diagnosed at any age are considered to have familial prostate cancer [15]. An earlier investigation conducted as part of the Health Professionals Follow-up Study until 1996 revealed that African American men experienced an 81% high chance of prostate cancer than White men, even after considering racial discrepancies in demographic variables [16]. For all first-degree family members (parents and children) and their second-degree family members (aunts and uncles grandparents, grandchildren, nieces, nephews, half-siblings, and double cousins), a family history for all types of malignancies including prostate cancer, was obtained [17]. It is widely acknowledged that, along with breast, ovarian, lung, and colorectal cancers, prostate cancer represents one of the major malignancies that is most frequently reported in families [18]. The only known risk factors for prostate cancer are, black ethnicity, old age and a family history of the condition [19]. Prostate cancer development and progression are influenced by both hereditary and environmental factors. The E26 transformation-specific (ETS) family members, SPOP (Speckle-type poz protein), FOXA1 (Forkhead box protein A1), IDH1 (isocitrate dehydrogenase 1), TP53 (tumor protein 15), RBL, and phosphatase and tensin homolog (PTEN) are among the genetic factors that may be involved [20]. Early onset is a hallmark of hereditary prostate cancer, which is generally autosomal dominant inheritance. In the study conducted by Heidegger

et al. [21], males diagnosed under 55 years of age account for 43% of the cumulative fraction of prostate cancer cases attributed to high-risk susceptibility alleles, whereas men beyond 85 years of age only account for 9%. Alternatively, at a single chromosomal region at 8q24, a variety of single nucleotide polypeptides have been associated to prostate cancer risk across different ethnic groups [22]. There is evidence from epidemiological and observational research that improper diets, excessive consumption of alcohol, and usage of tobacco are linked to an increased risk of chronic non-communicable illnesses, such as malignancies [23]. The likelihood of dying from prostate cancer and recurrence is higher in individuals who are obese. An increase in a body mass index of 5 kg/m² was linked to a 20% increased risk of mortality from prostate cancer in a meta-analysis of 6 post-diagnosis studies including men [24]. Between 2006 and 2020, a total of five studies were published, three of which looked at recurrence or advancement, and all five explored the relationship between post-diagnostic fish intake on prostate cancer outcomes. There were no correlations found between the overall number of poultry or the total amount of processed or unprocessed red meat in any of the three studies that looked at post-diagnostic diet of meat, chicken, and eggs and recurrence/progression between 2006 and 2016 [10]. In analyzing the mortality specific to prostate cancer, the two research studies concurred that there was no correlation between the general trend of liquor or alcohol intake. But in studies that did not include non-drinkers, the Canadian study found a positive correlation with moderate alcohol use [25]. The two primary objectives of the current prostate cancer early detection programs are: (i) recognizing non-symptomatic individuals with Prostate cancer requiring prostatic core biopsy; and (ii) identify men with prostate cancer who might benefit from definitive medical treatment [26]. Following the implementation of prostate specific antigen (PSA) screening over thirty years, there has been a notable decline in the frequency of metastatic illness and a decrease in the death rate specifically related to prostate cancer [27]. PSA is the only biomarker for prostate cancer diagnosis and prognosis that has been authorized by the U.S. Food and Drug Administration (FDA) [28]. Serum PSA levels being elevated may indicate a sign of prostate cancer; however, PSA is not a very precise screening tool since elevated serum PSA levels are not only associated with prostate cancer [29]. PSAs and rectal examinations are two screening methods for prostate cancer. A prostate biopsy is often initiated in case of abnormal rectal examination results or a significantly elevated PSA level. The PSA's performance is not ideal; depending on the cut off applied, only 23–30 and 19%–26% of biopsy re-

sults will identify any prostate cancer or clinically relevant prostate cancer (Gleason score 7 or above), respectively [30]. The identification of early biomarkers for the advancement of prostate cancer is essential for estimating the likelihood of recurrence and limiting the use of PSA for active surveillance. Some of the available biomarkers include tissue-based studies, serum assessment for derivatives of PSA, and pre-diagnostic urine-based examination [31]. In liquid biopsies, extracellular vesicles (EV) provide a supplementary source of cancer-derived molecules. All lipid bilayer particles that are spontaneously released from cells which are incapable of replicating are referred as EVs [32]. Urine can be used to extract secreted prostate fluid which is a potentially valuable source of biomarkers, without intrusive methods. Prostate secretion's composition has been shown to reflect the health of the prostate gland and the prostate tissue, providing a window for cancer detection, as changes in the function and makeup of the secretions are linked to the development of the disease [33]. On evaluating the end products of metabolic events from cells and tissues, metabolomic profiling of cancer offers insight into its current physiological and pathophysiological state, making it a promising technique for discovering biomarkers [34]. The 4Kscore is a clinical-based predictive tool for prostate cancer that is based on the assessment of four kallikrein proteins (total PSA [tPSA], free PSA [fPSA], intact PSA, and human glandular kallikrein-2 [hK2]) in blood in addition to clinical factors such as age, results from digital rectal examination (DRE), and prostate cancer history [35]. A numerical equation based on the concentrations of three PSA isoforms (tPSA, fPSA, and proPSA) is referred as the prostate health index (PHI) [36]. Large chain RNA molecule known as prostate cancer antigen 3 (PCA3) is overexpressed in malignant prostate tissue when compared to benign tissue which is located on chromosome 9q21–22 inside the intron 6 of the *PRUNE2* gene [37]. The Genomic Prostate Score (GPS) is determined by assessing the expression of five reference genes and 12 genes associated with the severity of prostate cancer [38]. The genes associated with cancer that have been chosen for this genomic test include those that are involved in stromal response (COL1A1, SFRP4, and BGN), *Xenopus* kinesin-like protein 2 (TPX2), androgen pathway (RAM13C, KLK2, AZGP1, and SRD5A2), and cellular organization (GSTM2, GSN, FLNC, and TPM2) [39]. Another polygenic genomic test for cancer risk stratification is the Prolaris Score. This test was created by Myriad Genomics for examining tissue samples embedded in formalin and fixed in paraffin after biopsy or prostatectomy [40]. In conclusion, a recently validated test for prostate cancer prognosis using eight biomarker multiplex

immunofluorescence based assay utilizes prostate core biopsy material [26]. The biomarker panel consists of the eight biomarkers which includes: GOLM-1 (Golgi membrane protein 1), PEDF (pigment epithelial-derived factor), serum CD14 (monocyte differentiation antigen CD14), IGFBP-3 (insulin-like growth factor binding protein 3), ETS-related gene (ERG) protein, IGF-1 (insulin like growth factor-1), VEGF-D (vascular endothelial growth factor-D) including PSA as the benchmark, which has been addressed according to the current diagnostic biomarkers for prostate cancer [41].

BIOMARKERS

Biomarkers for diagnosis, prognosis, and prediction facilitate patient-specific treatment of prostate cancer. Serum PSA testing, DRE, and systematic transrectal ultrasonography-guided biopsies are the core components for the management of prostate cancer [42]. PSA is generally a serine protease, which is made up of 261-amino acid protein that contains a 17-amino acids signal peptide followed by a 7-amino acids pro-peptide. Because of its increased levels in the serum and prostate, PSA is a sensitive diagnosis for prostate cancer. Both complex and fPSA forms are included in serum tPSA. Another important PSA type is PSA-ACT (α -1-antichymotrypsin) complex, which rises when PSA concentration increases [43]. Various extremely small-volume tumors with PSA levels between 4 ng/mL and 10 ng/mL have been found as a result of the wide usage of PSA testing. With these results, the probability of a biopsy-based diagnosis of prostate cancer is 25%. Additionally, PSA testing lacks the ability to distinguish between tumors that are highly and less aggressive, increasing the risk of overdiagnosis and unnecessary biopsies [44]. To assist the prediction of aggressive cancer upon biopsy or repeated biopsy, several fluid-based molecular biomarkers have been developed, including the PHI, ProgenSA, 4Kscore, MyProstateScore, Select mdx, ExoDX Prostate, Apify, and Proclarix [45]. The PCA3 test is used in the PCA3 biomarkers testing to quantify PCA3 mRNA in first-void and post-DRE urine. In addition to age and results from the DRE, the 4Kscore incorporates four distinct kallikrein indicators (tPSA, fPSA, intact PSA, and hK2) [46]. The analysis of RNA expression of 31 cell cycle progression (CCP) genes normalized to 15 housekeeping genes yields the CCP score. Earlier studies have confirmed that the score accurately predicts long-term mortality outcomes as well as post-prostatectomy recurrence [47]. Since a decrease in the percentage of α 2,6-sialic acid (SA) is linked to prostate cancer aggressiveness, aberrant glycosylation plays a role in the development of tumors and particular alterations in PSA glycosylation pattern.

Previous investigations characterized the primary sialylated PSA glycoforms from the blood serum of aggressive prostate cancer patients and compared them to standard PSA from the seminal plasma of healthy persons. Early research indicated that PSA's glycosylation might be a useful sign for a more accurate yet noninvasive prostate cancer diagnosis. Despite its low analytical sensitivity and specificity of the current assays, recent studies on the molecular features of PSA glycosylation have demonstrated its potential for improving prostate cancer diagnosis to a certain extent [48]. Recent studies indicate that exosomes produced from cancer have a role in the development of treatment resistance, and tumor progression [49]. Exosomes have been shown to be produced in large quantities by cancer cells, and these particles have the ability to alter recipient cells distantly, offering particular benefits in the microenvironment that aid in the growth and division of tumors. According to research by Hosseini-Beheshti et al. [50], exosomes formed from prostate cancer cells promote the growth and migration of RWPE-1 and LNCaP (lymph node carcinoma of prostate) cells by lowering apoptosis and transferring apoptosis inhibitory proteins [51]. Northern blot assay was employed in the identification of prostate transmembrane protein, androgen induced 1 (PMEPA1) mRNA, which is typically found in hormone-dependent tissues such as prostate and ovary. Further evidence from *in situ* hybridization revealed that benign prostate glandular epithelial cells were the primary expression site for PMEPA1

isoforms (a and b). Multiple solid tumors, including pancreatic endometrial tumor, prostatic adenocarcinomas, stomach cancer, renal cancer, breast cancer, and colon cancer, were shown to overexpress PMEPA1 (isoform - a) [52]. PTEN, Ki67, zinc-alpha 2 glycoprotein (AZGP1), p53 nuclear accumulation, and indicators of neuroendocrine development are some of the immunohistochemistry markers linked to metastases free survival and prostate cancer-specific mortality [53]. Using ELISA, the levels of transforming growth factor 1 (TGF1) and interleukin 6 (IL6) in the bloodstream were measured. The preoperative plasma levels of PF4 (platelet factor 4), IL6-SR (soluble receptor), and TGFβ1 were measured [54]. Since the non-invasive tissue biopsy enhances the awareness of treatment resistance profiles, metastatic potential, personalized therapeutics, and patient prognosis, genomic profiling via plasma DNA or circulating tumor cells (CTCs) can therefore play a critical role in guiding clinical decision making [55] (Fig. 1).

GOLD STANDARD TECHNIQUES

At present, PSA is recognized as a key tumor marker in evaluating the efficacy of therapy for individuals suffering from prostate cancer, prognostic factors, tumor mass assessment, early recurrence identification, and making it is an invaluable tool for prostate cancer screening and early diagnosis [56]. PSA is a serine protease related to kallikrein

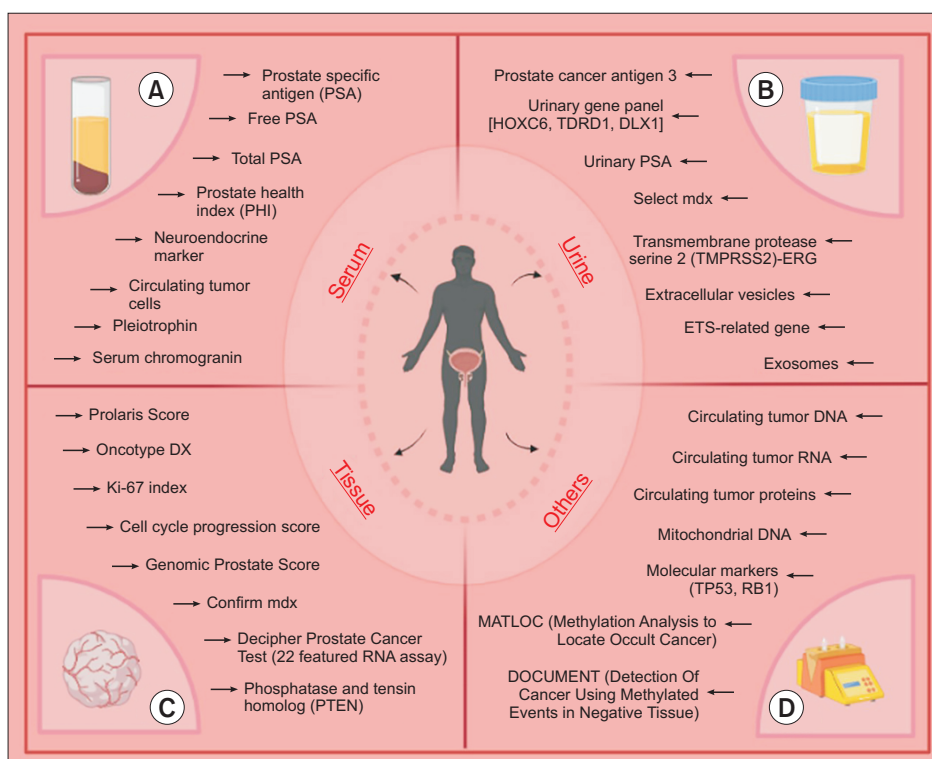


Fig. 1. Classification of biomarkers in the detection of prostate cancer using different samples. ERG, ETS-related gene; ETS, E26 transformation-specific.

generated by prostatic epithelial cells which generally occurs in prostate secretion which is frequently increased in the case of prostate cancer [57]. PSA includes three main forms, ACT-bound PSA makes up the majority of immunoreactive PSA in serum, accounting for around 75% of all PSA in circulation. A commercial test cannot identify the less than 0.1% of serum that is linked to alpha-2 macroglobulin by PSA, whereas 5%–50% of serum contains the free (enzymatically inactive) form of PSA [56]. While a minor portion of circulating PSA is free, its major portion is linked to proteins, primarily ACT. However, examining the ratio of fPSA versus tPSA can improve the diagnostic accuracy of PSA since a low proportion of fPSA, or a low f/t PSA, indicates significant prostate cancer, an association that was discovered shortly after the PSA test was developed [58]. PSA/tPSA ratio, or percent fPSA, was one of the first auxiliary tests created to improve PSA specificity. While there is no proven therapeutic benefit of measuring the absolute level of fPSA in blood, calculating the percentage of fPSA has been demonstrated to improve PSA's specificity in detecting prostate cancer, particularly in men whose tPSA levels fall between 2 µg/L and 4 µg/L and 10 µg/L [59]. Two forms of PSA are found in peripheral blood which includes an active form that is catalyzed and complexed to protease inhibitors, mostly ACT, and an inactive form which is not catalytically complexable rather circulates as fPSA [60]. Multiple proPSA isoforms were found during recent investigation of the fractions of fPSA. It has been determined that the peripheral region of the prostate gland is more specific for prostate cancer due to the elevated levels of four distinct truncated isoforms of proPSA [61]. PSA screening appears to enhance the discovery of prostate cancer at any stage, improves the detection of prostate cancer at stages I and II, and slightly reduce the detection of prostate cancer at stages III and IV, according to the evidence of moderate and poor quality. Simultaneously, it has no impact on overall mortality but may slightly lower the mortality specific to prostate cancer [62]. A transrectal core-needle biopsy guided by ultrasonography is usually performed on men whose PSA levels are consistently raised in order to screen for prostate cancer. Compared to the general population, PSA screening likely significantly enhances the identification of cancer in any stage in the individuals with prostate cancer [63]. The ratio of unbound PSA to the total quantity of PSA (all different forms) can be utilized to increase the diagnostic and prognostic utility of PSA in distinguishing benign prostatic problems from prostate cancer and predicting cancer risk. Thus, a lower fPSA level than the 15%–25% threshold is linked to a higher risk of prostate cancer; nevertheless, the diagnostic

cut off value for fPSA might vary depending on the doctors, the patient's age, and their clinical history [64]. The “prostate health index” or PHI, was developed using p2PSA as the base. PHI is a formula that combines p2PSA, fPSA, and tPSA. Studies has indicated that PHI increases the diagnosis accuracy for prostate cancer [65].

SERUM-BASED BIOMARKERS

The primary objective of the blood based liquid biopsies for the diagnosis and treatment of prostate cancer is to reduce the number of men who have needless biopsies, distinguish between indolent and clinically relevant prostate cancer, and prevent needless low-risk tumor diagnosis. Methylation-based indicators offer significant predictive and prognostic ability in addition to prostate cancer diagnosis [66]. Many of the novel serum indicators being studied at the moment involve tPSA or fPSA in the evaluation and interpretation of the data, raising the issue of whether these new markers possess the same limitations as PSA [67]. A serine protease found in prostate tissue is known as prostate-specific antigen, or PSA, which is a 240 amino acid glycoprotein that is present in both benign and cancerous prostate tissue which usually dissolves the semen coagulum and gets secreted in the semen. PSA is also present in amniotic fluid, periurethral glands, cystic breast fluid, liver, kidney, and lung tumors [68]. The PHI serves as a cheap and easy blood test that may be used in a multivariable screening strategy. It has demonstrated strong performance in this regard for prostate cancer diagnosis [69]. The PHI score exhibits a high rate of diagnostic accuracy and might be a valuable tumor marker in identifying individuals with more aggressive illness. Additionally, PHI forecasts the chance of progression under active observation [70]. PHI score is comprised of proPSA, fPSA, and tPSA. Perhaps these several PSA forms may be used in conjunction with one another for better identification of prostate cancer at high risk has been evaluated. The α 2,3-SA test is used to identify PSA glycoforms that include α 2,3-SA and have been associated with aggressive prostate cancer [71]. Based on the four kallikrein proteins (tPSA, fPSA, intact PSA, and hK2), the 4Kscore is generated. Since 2015, it has been included in the guidelines of National Comprehensive Cancer Network (NCCN) prostate cancer early detection [72]. The use of CTCs as biomarkers among individuals with metastatic prostate cancer has been extensively studied. Tumor cells have been found in both the bone marrow and bloodstream of men with prostate cancer, according to several investigations. The development of next-generation sequencing (NGS) and

CTC identification techniques in cancer patient's serum and plasma has led to the emergence of CTCs & circulating free tumor DNA as potential liquid biopsy tests in oncology [73]. Neuroendocrine marker (NEM) is a 38 kDa immunoreactive protein which was exclusively found in the cancerous—non benign regions of the prostate gland. Serum NEM range in prostate cancer Individuals were about three times larger than in individuals suffering from benign urologic illnesses, according to a preliminary analysis of 69 cancer and non-cancer patients. According to the findings, NEM was a more dependable method than PSA for accurately diagnosing prostate cancer patients [74]. Pleiotrophin (PTN) is a potential early indicator of pro-metastatic illness, and it may be especially important in men with favorable clinic-pathological traits. According to various studies, overexpressing PTN causes an aggressive phenotype, whereas downregulating PTN expression reduces the proliferation rate of prostate cancer cells, motility, and *in vitro* invasion as well as *in vivo* tumor development and its metastases [75]. In addition, four-biomarker panel of PHI, Fuc-PSA, SDC1, and GDF-15 or phi, Fuc-PSA, SDC1, and Tie-2 were found to perform well better than phi alone in separating aggressive prostate cancer from low risk prostate cancer or statistical significance from a low-risk prostate cancer and non-prostate cancer [76] (Fig. 2).

URINE-BASED BIOMARKERS

Urine offers several benefits over other biological specimens especially when it comes to the identification of prognostic and diagnostic biomarkers [77]. Prostate gland secretes components that may be transported to the urinary system and subsequently identified in urine, these factors can serve as biological markers for the detection and prognosis of prostate cancer [78]. Patients with prostate cancer have lower urinary PSA than men without cancer symptoms, and this decreases gradually as tumors stages increases. Significantly, there were no changes found between healthy people and individuals with low-risk tumors, indicating that it may be possible to distinguish between prostate cancer being indolent and clinically significant [79]. The transcript of transmembrane protease serine 2 (TMPRSS2)-ERG gene fusion is another biomarker that is highly correlated with malignant prostatic epithelial cells. This androgen-regulated gene fusion is lacking in specimens which don't have prostate cancer and is present in almost 50% of individuals with prostate cancer [80]. The fusion gene ERG (TMPRSS2-ERG), which is often measured by reverse transcriptase polymerase chain reaction (RT-PCR), is a reliable biomarker for prostate cancer. An MS (MRM)-based assay developed by He et al. [81] made it feasible to analyze the ERG protein urinary, which exhibited excellent sensitivity and accuracy. PCA3 and Select mdx are the two urine biomarkers, demonstrated the ability

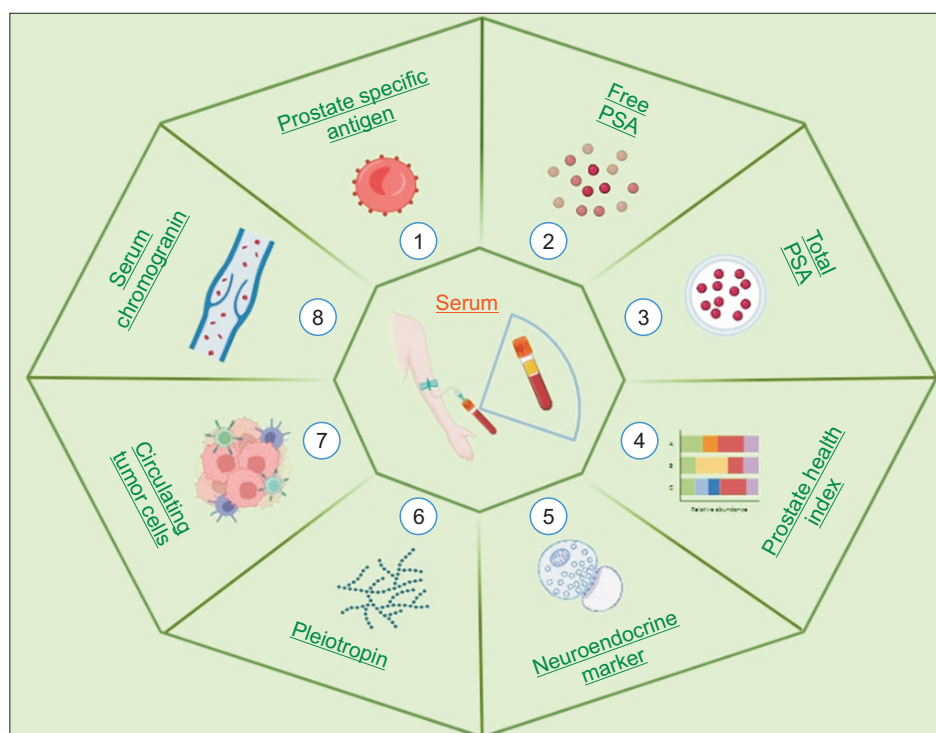


Fig. 2. Serum-based biomarkers used in diagnosis of prostate cancer. PSA, prostate specific antigen.

to increase prostate cancer detection specificity. These tests discover a lot of malignancies that might not be harmful if left undetected, while being a major advancement over PSA alone [82]. Notably, exosomes have been suggested as a source of biomarkers as well as biomarkers themselves, for the purpose of tracking patients undergoing cancer therapy [83]. A new method being developed to increase the specificity and sensitivity of biomarkers for the identification and evaluation of prostate cancer is the isolation of EVs from urine. EVs are known to be abundant sources of biomarkers; however, their application in diagnosis has been hampered by the lack of clinically feasible techniques for separating them from biofluids [84]. A molecule found on the cell membrane called CD90 has been found to be elevated in a number of malignancies, including prostate cancer. Tests evaluating the presence of CD90 fragments in patient's voided urine before and after prostatectomy revealed that a CD90 peptide was specifically identified in patients before the procedure [85] (Fig. 3).

TISSUE-BASED BIOMARKERS

The Oncotype DX genomic test is a biopsy-based assay that measures 17 gene expressions in RNA samples collected from the tumor tissue that has been fixed from prostatic needle biopsies, which is based on RT-PCR and has

been statistically validated. On a scale of 0 to 100, the assay yields a GPS where results with higher values suggest more biologically aggressive illness. It has been clinically demonstrated as a consistent and significant indicator for adverse pathology (AP) and biochemical recurrence after radical prostatectomy in men with clinically very low-, low-, and intermediate-risk of prostate cancer [86]. The gene expression levels linked with the 4 tumor aggression pathways such as androgen signaling (AZGP1, KLK2, SRD5A2, and FAM13C), cellular organization (FLNC, GSN, TPM2, and GSTM2), stromal response (BGN, COL1A1, and SFRP4), and cellular proliferation (TPX2)—are measured in this test using reverse transcription quantitative polymerase chain reaction (RT-qPCR) [87]. The researchers concluded that FUT8 might have potential as a predictive biomarker since it has been found to be overexpressed in metastatic tissue compared to normal tissue and to be increased in metastatic tissue compared to normal tissue [88]. The Decipher Prostate Cancer Test is an RNA biomarker test with 22 features designed to predict the probability of metastases at 5 years and 10 years following radical prostatectomy (RP). A total of 22 coding and non-coding genes are involved in seven cancer pathways, including vascular development, metastasis and invasion, growth and differentiation are included in this assay, which produces a score between 0 and 1, where greater values signify a higher likelihood of both adverse patient

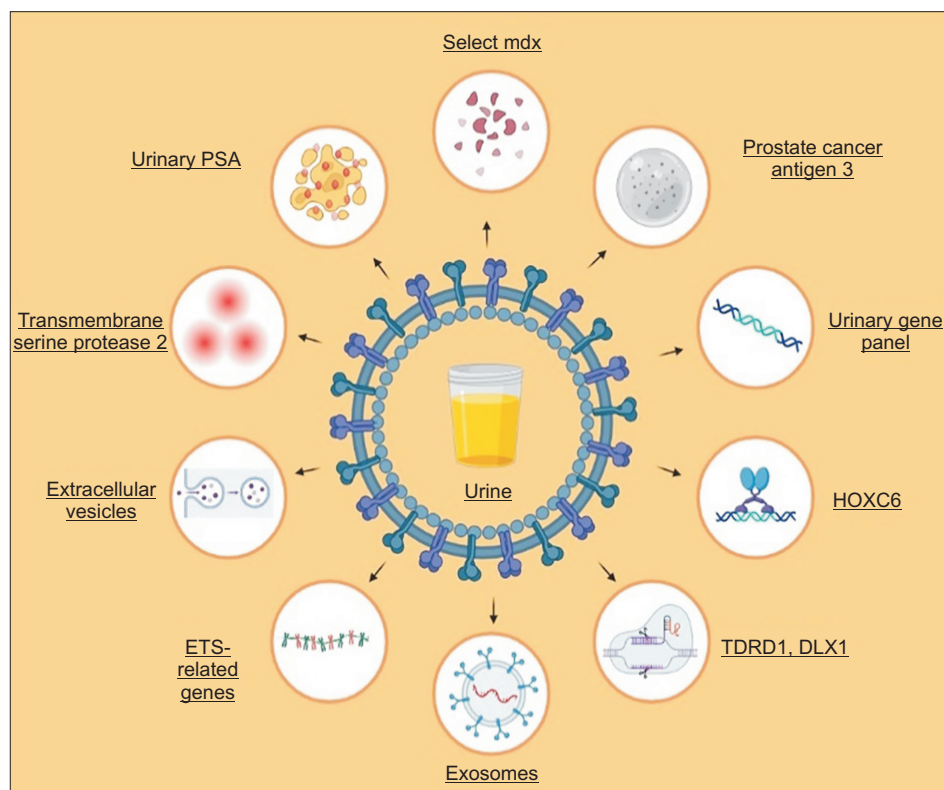


Fig. 3. Urine-based biomarkers used in diagnosis of prostate cancer. PSA, prostate specific antigen; ETS, E26 transformation-specific.

outcomes (AP) and inferior oncologic outcomes [89]. The CCP is a well-established predictive RNA expression profile that utilizes the gene expression levels of fifteen housekeeping genes and other 31 genes associated with CCP. Numerous research investigations have indicated a high correlation between the CCP score and the development of metastatic illness following both surgery and radiotherapy [90]. Dysregulation of PTEN appears to be more common than other mutations in advanced, localized, or metastatic prostate cancer and has been demonstrated to have an effect on prognosis, taking into account variations that exhibit a well-found involvement in prostate cancer. PTEN is a tumor suppressor gene which is found in chromosome number 10 whose mutation is known to be a component in PI3K/AKT signaling cascade. Deletions of PTEN have been found in 10%–70% of RP cases. PTEN tends to be the most commonly altered tumor suppressor gene; the heterogeneity in PTEN deletion rates is likely connected to the risk group, where higher risk corresponds to a higher incidence of PTEN deletion. A trans-membrane glycoprotein called prostate-specific membrane antigen (PSMA) is overexpressed in prostate cancer. With their exceptional ability to bind to the active extracellular center of prostate cancer, radio-labeled small molecules have become a promising new standard of reference for prostate cancer diagnosis, producing images with remarkable tumor-to-background contrast. In non-prostatic tissues such the lacrimal gland, neurological system, duodenum, and normal prostatic tissues, PSMA expression is incredibly low. On the other hand, compared to normal tissues, its expression is multiplied 100–1,000 times in prostate cancer tissues. In par-

ticular, the expression level is significantly higher in prostate cancer tissues that are undifferentiated, metastatic, and resistant to castration. Although PSMA positron emission tomography (PET) is often utilized for theranostic selection in metastatic prostate cancer, its potential utility for therapeutic monitoring remains unclear. However, there is a lack of data about its appropriate application to enhance patient-related outcomes, particularly when oligometastatic illness is targeted for treatment and there is an early biochemical recurrence [91]. Two retrospective patient cohorts have been used to evaluate the usefulness of determining the cancer on follow-up biopsies following an initial negative biopsy: (i) the development of the MATLOC (Methylation Analysis to Locate Occult Cancer) in UK and Belgium, which evaluated 483 subjects, and (ii) the DOCUMENT (Detection of Cancer Using Methylated Events in a negative Tissue) [92]. Prominent research has demonstrated the growing significance of magnetic resonance imaging (MRI) in the detection of prostate cancer and emphasized its possible advantages. MRI can be employed for two purposes: first, in the pre-biopsy context, to determine if a biopsy is necessary as per the presence or absence of doubtful lesions; and second, to focus biopsies on those lesions. Consequently, a new paradigm for the screening of prostate cancer may be established by combining enhanced risk prediction with MRI-targeted biopsy [93]. In addition to these licensed substances, the use of PSMA-based PET-CT (computed tomography) and PET-MRI, which detects metastases in regional lymph nodes and is particularly useful in patients with low PSA levels, compares favorably with current modalities [94] (Fig. 4).

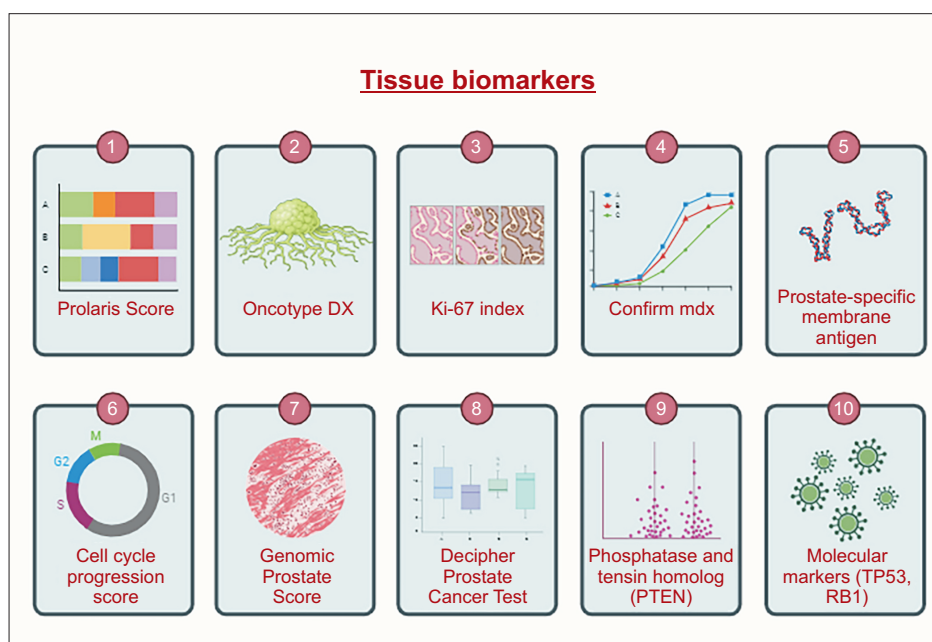


Fig. 4. Tissue-based biomarkers used in diagnosis of prostate cancer.

BIOMARKERS USED IN CLINICS

Prolaris, Decipher, Oncotype DX Prostate, and ProMark are some of the examples of commercially accessible molecular biomarkers which are made available in circumstances when the assay result taken into account in conjunction with standard clinical criteria, which is expected to have an impact on management [4]. The most frequently identified genomic abnormalities in prostate cancer are fusions of the ETS gene, androgen receptor, PTEN, c-MYC, and miRNA profiles. These variants are being investigated as potential therapeutic targets as well as biological markers for disease stratification and resistance to therapy [7]. Primary prostate cancer often exhibits changes in PTEN, which are enriched in metastatic castration-resistant prostate cancer. Additionally, PTEN is frequently altered, and the kinases PI3K and AKT are activated downstream of one another [95]. Currently available methods such as RT-qPCR, enzymatic luminescent miRNA test, microarrays, *in situ* hybridization or NGS should be used to identify the tumor at the earliest stage feasible [96]. Compared to benign prostatic hyperplasia, the synthesis of the calcium-binding urine protein ANXA3 is markedly reduced in prostate cancer. Antibodies against the HER-2 (human epidermal growth factor receptor-2)/neu, prostatic acid phosphatase, p53, glucose-regulated protein 78, AMACR (alpha-methylacyl-CoA racemase), and PSA have

been identified in prostate cancer individuals [97] (Table 1).

CONCLUSIONS

In summary, recent research has validated the essential function of biomarkers in the diagnosis and treatment of prostate cancer. The identification of clinically relevant prostate biopsies has been accomplished using a number of intriguing molecular biomarkers; yet, no firm findings about biomarker superiority could be drawn. The rapid prevention of metastatic illness and the selection of appropriate therapy may also benefit from the integration of clinically relevant prognostic and predictive biomarkers. More targeted therapy for patients with higher-risk circumstances, overtreatment, and improved risk appraisal comprise all possible advantages of the recently identified, more precise prostate cancer biomarkers.

CONFLICTS OF INTEREST

The authors have nothing to disclose.

FUNDING

This study was funded by Department of Science and Technology, Core Research Grant (grant number:

Table 1. FDA and CLIA approved serum-, urine-, and tissue-based biomarkers for prostate cancer detection

Specimen	Test	Biomarker	Approved by
Serum	PSA test	PSA	FDA
	4Kscore test	Total PSA, free PSA, intact PSA, hK2	CLIA
	PHI	Total PSA, free PSA, proPSA (p2PSA)	FDA
	Neuroendocrine markers	Chromogranin A, androgen receptor, synaptophysin, neuron-specific enolase	FDA
Urine	Progenesa score	PCA3	FDA
	Select mdx	Urinary gene panel (HOXC6, TDRD1, DLX1)	CLIA
	MyProstateScore	PCA3, PSA, TMPRSS2-ERG	CLIA
	ExoDX Prostate score	Exosomal RNA	CLIA
Tissue	Prolaris Score	mRNA expression of 31 genes	FDA
	Decipher Prostate Cancer Test	22 gene featured RNA assay	CLIA
	ProMark assay	8 gene panel assay: VEGF-D, ERG, IGF-1, IGFBP-3, GOLM-1, PEDF, serum CD14, PSA	CLIA
	Oncotype DX	mRNA expression of 17 genes	CLIA
	Confirm mdx	Hypermethylation of DNA	CLIA
	Ki-67 index	Cell proliferation marker	FDA

FDA, U.S. Food and Drug Administration; CLIA, Clinical Laboratory Improvement Amendments; PSA, prostate specific antigen; hK2, human glandular kallikrein-2; PHI, prostate health index; PCA3, prostate cancer antigen 3; TMPRSS2, transmembrane protease serine 2; ERG, ETS (E26 transformation-specific)-related gene; VEGF-D, vascular endothelial growth factor-D; IGF-1, insulin like growth factor-1; IGFBP-3, insulin-like growth factor binding protein 3; GOLM-1, Golgi membrane protein 1; PEDF, pigment epithelial-derived factor; CD14, monocyte differentiation antigen CD14.

2021/001369).

AUTHORS' CONTRIBUTIONS

Research conception and design: Vinoth-Kumar Lakshmanan. Data acquisition: Sathish Kumar AM and Prabhakar Rajan. Statistical analysis: Mohammad Alkhamees and Merrel Holley. Data analysis and interpretation: Vinoth-Kumar Lakshmanan, Sathish Kumar AM, Prabhakar Rajan, Mohammad Alkhamees, and Merrel Holley. Drafting of the manuscript: Sathish Kumar AM and Vinoth-Kumar Lakshmanan. Critical revision of the manuscript: Prabhakar Rajan, Mohammad Alkhamees, and Merrel Holley. Obtaining funding: Vinoth-Kumar Lakshmanan. Administrative, technical, or material support: Vinoth-Kumar Lakshmanan. Supervision: Vinoth-Kumar Lakshmanan, Prabhakar Rajan, Mohammad Alkhamees, and Merrel Holley. Approval of the final manuscript: all authors.

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